

HAT-P-36b

R-1132

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_220426 · created 2026-07-30T23:06:32+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459695.8928 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.099 +/- 0.035
Transit depth (Rp/Rs)^2	0.97 +/- 0.69 [%]
Orbital Inclination (inc)	88.73 +/- 2.81
Airmass coefficient 1 (a1)	1.393 +/- 0.022
Airmass coefficient 2 (a2)	-0.1659 +/- 0.0089
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	None
Optimal Aperture	3.8
Optimal Annulus	13.99
Transit Duration (day)	0.0932 +/- 0.0046

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.099 ± 0.035 vs 0.126 (deviation 0.77σ)
Duration fit vs published	0.0932 d vs 0.0925 d (deviation 0.15σ)
Mid-time O–C	10.9 min
Depth SNR	2.8
Residual scatter	1.53 %

Light curve

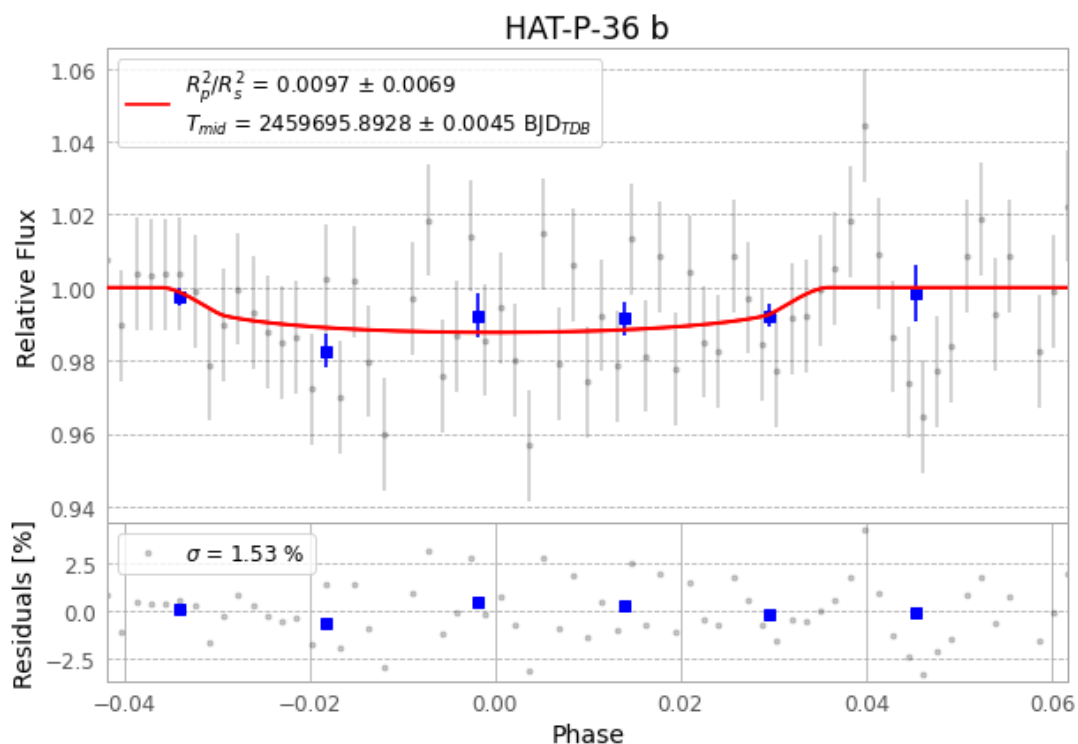


Figure 1 — FinalLightCurve_HAT-P-36 b_2022-04-26.png (SHA-256 97e5b8bf7f6df7d7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [278, 215], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 44d17a64d078b632...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), HATP-36220426080014.FITS → HATP-36220426111814.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-220426.log (7171efc875af...), FinalLightCurve_HAT-P-36 b_2022-04-26.png (97e5b8bf7f6d...), FinalLightCurve_HAT-P-36 b_2022-04-26.csv (204ccb3f96eb...)

Compute resources

Wall time 115.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 23:06Z.